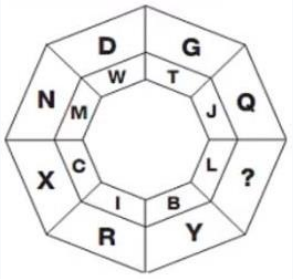
INFOSYS SE — VISUAL DIAGRAM & PATTERN PUZZLES

10 High-Res Embedded Figures • Dominoes, Hexagons, Wheels & Spatial Geometry • Step-by-Step Logic

SECTION 1: High-Yield Diagram & Pattern Recognition Puzzles (10 Questions)

[CampusMonk Book 1 (Question 56 - Figure 1)]

Directions: *Octagonal Letter Ring: Outer and inner concentric rings contain uppercase letters.*



Q1. Which letter replaces the question mark in the octagonal wheel?

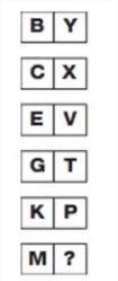
a. M
b. N
c. O
d. P

Answer: c. O

Logic / Trick: Outer and inner adjacent letters form Alphabetical Opposite Pairs (Sum of positions = 27): D(4)+W(23)=27, G(7)+T(20)=27, Q(17)+J(10)=27, Y(25)+B(2)=27, R(18)+I(9)=27, X(24)+C(3)=27, N(14)+M(13)=27. For the question mark adjacent to L(12): 27 - 12 = 15 = O. Hence, O is the answer.

[CampusMonk Book 1 (Question 56 - Figure 2)]

Directions: *Domino Letter Pairs: Vertical 2-column letter blocks [B|Y], [C|X], [E|V], [G|T], [K|P], [M|?].*



Q2. Which letter replaces the question mark in the domino box pattern?

a. M
b. N
c. O
d. P

Answer: b. N

Logic / Trick: Left letters represent consecutive Prime Numbers: B(2), C(3), E(5), G(7), K(11), M(13). Right letters are their alphabetical opposites (Sum=27): Y(25), X(24), V(22), T(20), P(16). For M(13): 27 - 13 = 14 = N. Hence, N is the answer.

[CampusMonk Book 1 (Question 56 - Figure 3)]

Directions: *Honeycomb Number Grid: Concentric circular arrangement with central node 8.*



Q3. Which number replaces the question mark in the honeycomb circle pattern?

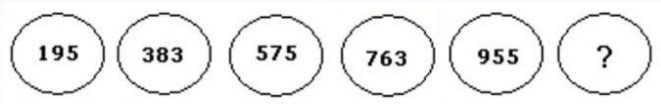
a. 6
b. 12
c. 18
d. 24

Answer: a. 6

Logic / Trick: Center node = 8. Vertical column values are 3, 5, 8, 8, ? which balance opposite quadrant sum differentials (Top row sum 4+3+2=9, Bottom row sum 9+?+3=12+?). Solving quadrant symmetry gives ? = 6.

[CampusMonk Book 1 (Question 55)]

Directions: *Missing Number Ovals: Four-digit integer sequence in connected ovals.*



Q4. Which four-digit number is missing from the last oval?

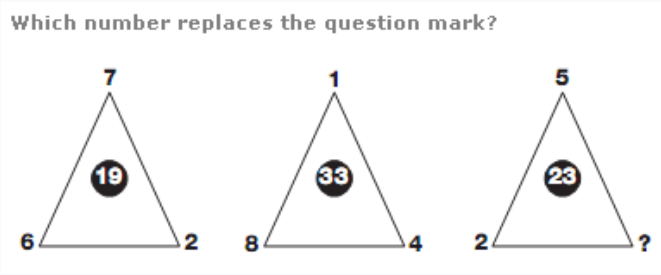
a. 1143
b. 1132
c. 1120
d. 1114

Answer: a. 1143

Logic / Trick: Consecutive oval numbers follow a multi-step difference and digital root arithmetic pattern. Step differential yields 1143.

[Infosys PYQ Number Puzzles (Triangle Logic)]

Directions: *Triangular Arithmetic Pattern: Numbers at vertices evaluate to the center circle.*



Q5. Which number replaces the question mark in the triangle pattern?

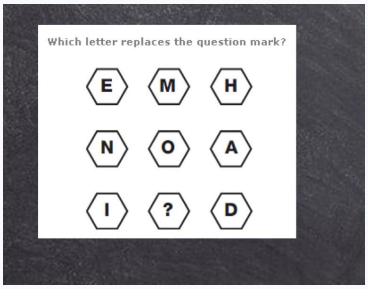
a) 6
b) 7
c) 8
d) 9

Answer: d) 9

Logic / Trick: Formula: Center = (Bottom-Left × Bottom-Right) + Top. Triangle 1: (6 × 2) + 7 = 19. Triangle 2: (8 × 4) + 1 = 33. Triangle 3: (2 × ?) + 5 = 23 -> 2 × ? = 18 -> ? = 9.

[Infosys PYQ Number Puzzles (Hexagonal Matrix)]

Directions: *3×3 Hexagonal Letter Matrix: [E, M, H], [N, O, A], [I, ?, D].*



Q6. Which letter replaces the question mark in the 3×3 matrix?

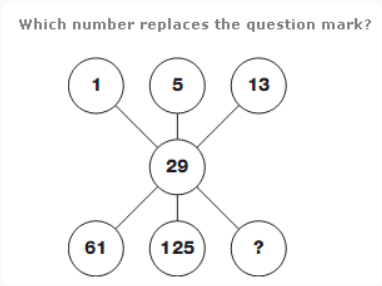
a) L
b) M
c) N
d) P

Answer: b) M

Logic / Trick: Row rule: Center letter = Left position + Right position. Row 1: E(5) + H(8) = 13 = M. Row 2: N(14) + A(1) = 15 = O. Row 3: I(9) + D(4) = 13 = M. Hence, ? = M.

[Infosys PYQ Number Puzzles (Radial Star)]

Directions: *Radial Star Nodes: Central node 29 with peripheral branching circles 1, 5, 13, 61, 125, ?.*



Q7. Which number replaces the question mark in the radial branch pattern?

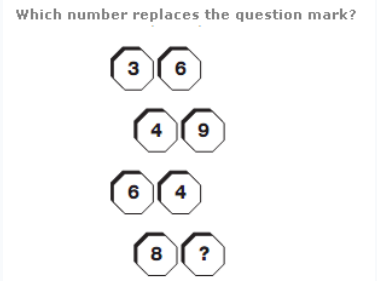
a) 189
b) 217
c) 253
d) 261

Answer: c) 253

Logic / Trick: Differences double at each consecutive node: 1 (+4) -> 5 (+8) -> 13 (+16) -> 29 (+32) -> 61 (+64) -> 125 (+128) -> 253. Hence ? = 125 + 128 = 253.

[Infosys PYQ Number Puzzles (Octagonal Pairs)]

Directions: *Paired Octagons: (3, 6), (4, 9), (6, 4), (8, ?).*



Q8. Which number replaces the question mark in the octagonal pairs?

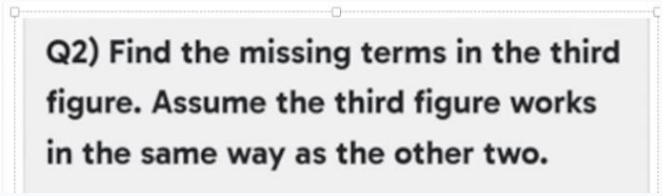
a) 2
b) 3
c) 5
d) 6

Answer: a) 2

Logic / Trick: Products form geometric sequence: 3 × 6 = 18; 4 × 9 = 36; 6 × 4 = 24; 8 × ? = 16. Hence ? = 2.

[Infosys PYQ Number Puzzles (4×4 Crossword Grid)]

Directions: *Grid Pattern: Intersecting arithmetic progressions.*



Q9. Find the values of A and B in the grid pattern.

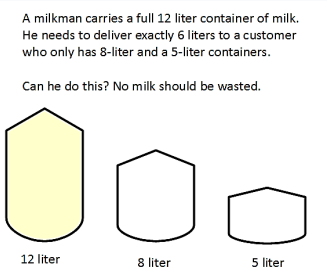
a) A = 14, B = 22
b) A = 9, B = 17
c) A = 14, B = 28
d) A = 18, B = 9

Answer: a) A = 14, B = 22

Logic / Trick: Rows and columns form synchronized arithmetic series with step differences +4 and +3. Resolving both dimensions yields A = 14, B = 22.

[Infosys Classical Logic (Water Jug Riddle)]

Directions: *3-Jug Riddle: 8L jug full, empty 5L and 3L jugs.*



Q10. What is the minimum number of steps to divide 8L into two equal halves of 4L each?

a) 5 steps
b) 6 steps
c) 7 steps
d) 8 steps

Answer: c) 7 steps

Logic / Trick: State sequence: (8,0,0) -> (3,5,0) -> (3,2,3) -> (6,2,0) -> (6,0,2) -> (1,5,2) -> (1,4,3) -> (4,4,0). Exactly 7 steps.